
An Analysis of Global and Regional Mainstreamness for Personalized Music Recommender Systems

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Abstract

The *music mainstreamness of a listener* reflects how strong a person's listening preferences correspond to those of the larger population. Considering that music mainstream may be defined from different perspectives, we show country-specific differences and study how taking into account music mainstreamness influences the quality of music recommendations.

In this paper, we first propose 11 novel mainstreamness measures characterizing music listeners, considering both a global and a country-specific basis for mainstreamness. To this end, we model *preference profiles* (as a vector over artists) for users, countries, and globally, incorporating artist frequency, listener frequency, and a newly proposed TF-IDF-inspired weighting function, which we call artist frequency-inverse listener frequency (AF-ILF). The resulting preference profile for each user u is then related to the respective country-specific and global preference profile using fraction-based approaches, symmetrized Kullback-Leibler divergence, and Kendall's τ rank correlation, in order to quantify u 's mainstreamness. Second, we detail country-specific peculiarities concerning what defines the countries' mainstream and discuss the proposed mainstreamness definitions. Third, we show that incorporating the proposed global and country-specific

mainstreaminess measures into the music recommendation process can notably improve accuracy of rating prediction.

Keywords: music mainstreaminess, music recommender systems, artist frequency-inverse listener frequency, popularity, country-specific differences.

1 Introduction

In the era of digitalization, music has become easier to access than ever: a tremendous number of musical recordings are readily available to consume on online platforms such as YouTube, Spotify, or iTunes. This opportunity to access a large number of musical works, though, results in information overload [8], which requires new tools to assist users in choosing from the huge amount of musical content [39]. Music recommender systems (MRS) have, thus, become a significant research topic over the past few years [6, 11, 43] and current online music platforms typically use some sort of MRS.

In general, the idea behind recommender systems is to assist users in searching, sorting, and filtering the vast amount of information available [29]. MRS are specifically built to assist users in navigating through the myriad of available musical recordings and provide them with music suggestions that would fit the respective user's interest or, respectively, automatically generate consecutive recommendations that build a personalized playlist [43]. The challenge is "to propose the right music, to the right user, at the right moment" [24].

Various automatic approaches to music recommendation have been proposed [45]. As summarized in the review by Schedl et al. [45], most MRS rely mainly on some sort of content-based filtering [5] or collaborative filtering [26]. Content-based MRS may, for instance, consider acoustic similarity information on the song level [49], or use the song's music genre, or the performing artist of the music item to quantify similarities [27]. MRS employing collaborative filtering do not require exogenous information about neither users nor music items. Instead, a user is suggested music listened to by users with similar preferences or listening patterns [34].

Another variant, popularity-based recommendation approaches, resemble a primitive form of collaborative filtering, where items are recommended to users based on how popular those items are overall among other users. Such approaches are built on the assumption that the target user is more likely to like

a very popular item than one of the far less popular items [11, 44]. Popularity-based recommendation approaches are particularly applicable in hit-driven domains—such as in the music industry. Accordingly, popularity-based MRS approaches are widely adopted to complement other approaches in cold start situations, when there is limited information about new users and/or items available in the system [13, 50].

One approach for considering popularity in the music domain is to describe music listeners “in terms of the degree to which they prefer music items that are currently popular or rather ignore such trends” [38]. Harnessing music mainstreamness in combination with collaborative filtering techniques tends to deliver better results with respect to music recommendation accuracy and rating prediction error than pure collaborative filtering approaches alone [16, 41, 44, 48].

However, a limitation of existing work on quantifying a user’s music mainstreamness is that music mainstream is viewed from a global perspective. There exist regional peculiarities to mainstream, though [7]. For instance, music consumption behavior is affected by culturally influenced music preferences, market regulations, local radio airplay, etc. (e.g., [10, 20, 35, 47]). In other words, regional aspects shape users’ music preferences and music consumption behavior. Accordingly, we can assume country-specific differences concerning which artists are popular.

With respect to the music recommendation research domain, the definition of specific measures that can capture a user’s mainstreamness (i) on both, a global and a country-specific level, and (ii) in ways that can easily be operationalized in music recommendation is a new target of research (e.g., [7, 41]). Calling on this, the main contributions of this paper are three-fold: (i) the definition of several novel measures for user mainstreamness, considering both a global and a regional, country-specific basis, (ii) the illustration of country-specific peculiarities of these mainstreamness definitions, and (iii) an analysis of the performance of the proposed mainstreamness measures for personalized music recommendation.

The remainder of the paper is organized as follows. In Section 2, we provide a brief overview over existing work on mainstreamness and popularity in music recommendation, and introduce the dataset on which we conduct our experiments. We then detail the proposed mainstreamness measures in Section 3 and provide examples that show their value to distill the regional mainstream, in addition to a global one. In Section 4, we discuss for a few prototype countries the relationship between their regional mainstream in comparison to the global mainstream. Section 5 shows how to exploit the

proposed mainstreamness measures in collaborative filtering recommendation and highlights the additional values of doing so. Eventually, we round off the paper in Section 6 with a conclusion and directions for future research.

2 Conceptual Foundations and Related Work

2.1 Music Popularity and Mainstreamness

In the context of recommender systems, popularity-based approaches are widely adopted in numerous domains, including music [13, 23, 50], news [51], or product recommendation in electronic commerce in general [1]. Popularity is thereby typically constructed as a general consensus of a group’s attitude about entities [23].

While various ways exist to define and measure popularity (for instance, in terms of sales figures, media coverage, etc.), in the field of MRS, music popularity is frequently characterized by using the total playcounts of a music item—i.e., the number of listening events the music item realizes by all listeners in total cf. [11]. With respect to music popularity by using playcounts, the *long tail concept* as described in [2] is specifically applicable to the (online) music industry [12]; on online music platforms there is a concentration of playcounts on the most popular music items (the head), and then there is a long tail of less popular items [9, 11].

A more general concept to popularity concentration is referred to as *mainstream*. Although literature in the field of popular music studies and popular music cultures references to *mainstream* frequently, the term itself remains rather poorly defined, cf. e.g., [4]. According to the Oxford Dictionaries, *mainstream* is defined as “The ideas, attitudes, or activities that are shared by most people and regarded as normal or conventional”. Due to the strong connection of the concepts, the terms *mainstream* and *long tail* are often used interchangeably. The mainstream is thereby frequently also referred to with other terms and phrases (e.g., *hits* [11], *the head* [15]) to circumscribe the phenomenon; the overall concept is also called, for instance, the hit-driven paradigm [11], the long-tail concept [2, 11], etc.

In MRS research, the user feature *music mainstreamness of a user* [16, 44] essentially describes whether and how strong a user’s music listening preferences correspond to those of the overall population. While other listening-centric features, for instance, serendipity [52] or novelty [14], are frequently exploited when modeling a user’s music consumption behavior and providing music recommendations, music mainstreamness is a rather

new target of research [16, 44, 48]. Thereby, the mainstreamness feature is used to analyze a user's ranking of music items and compare it with the overall ranking of artists, albums, or tracks [48].

2.2 Related Work on the Quantification of Music Mainstreamness

Formal definitions to measure the level of music mainstreamness of a user are scarce in literature (e.g., [41, 44, 48]). Most existing approaches quantify music mainstreamness as fractions of the target user's playcounts among the playcounts of the overall population. A limitation of this approach is that it disproportionately privileges the absolute top hits [41], which is problematic for long-tail distributions, which are present for music item popularity on online music platforms. There is a high concentration of demands on the most popular items and a long tail of less popular items. Privileging the top hits leads to low performance of fraction-based user models of mainstreamness in collaborative filtering approaches [41].

To overcome this limitation, Schedl and Bauer [41] proposed measurement approaches based on *rank-order correlation* and *Kullback-Leibler (KL) divergence*. However, also their work shares with existing fraction-based approaches to quantify mainstreamness that music mainstream is viewed from a global perspective and does not take regional peculiarities of music mainstream into account.

2.3 Cultural and Regional Aspects Influencing Music Mainstreamness

As human preferences and behavior are rooted and embodied in culture [22], also music preferences and music consumption behavior are affected by cultural aspects [17, 20, 47]. For instance, music perceptions vary across cultures [25, 30, 46, 47] and music preferences are shaped by cultural aspects [3]. For example, in the European countries, pop music preferences disconverge rather than converge [10].

Still, not only cultural aspects, but also regional (e.g., country-specific) mechanisms affect music consumption; particularly important are national market structures—including distribution channels, legislation, subsidizing, and local radio airplay—that vary across countries [19, 33, 35]. In other words, regional aspects shape users' music preferences and music consumption behavior. Being aware that culture does not equate nation [21, 28], we

emphasize that cultural aspects as well as national market structures contribute to users’ music consumption preferences and behavior. Accordingly, we can assume country-specific differences concerning the popularity of artists. Against this background, we focus on country-specific differences in the paper at hand.

Closest to our work is the study presented in [48], which analyzes the recommendation performance of mainstreamness (spelled “mainstreamness”) and a user’s country, among other features. Our work significantly differs from [48] in various regards: First, we use an open dataset to allow for replication. Second, [48] propose only one global mainstreamness measure that compares a user’s preferences to the overall dataset (global population), while we define mainstreamness in various ways (based on fractional, divergence, and rank correlation functions) and at various levels (global and country-specific). Third, we also propose a novel weighting approach based on “inverse listening frequency” that highlights artists popular in a specific country, thus, contributing to its mainstream, but not necessarily on a global level.

2.4 Data Preparation

For our experiments, we deploy the LFM-1b dataset [39], which covers 1,088,161,692 listening events of 120,322 unique users, who listened to 32,291,134 unique tracks by 3,190,371 unique artists. The core component of the dataset is the cleaned user-artist-playcount matrix (UAM) containing the number of listening events of 120,175 users to 585,095 unique artists. The distribution of listening events of the Last.fm data corresponds to a typical long-tail distribution [11]. As 65,132 user profiles do not contain any country information, we exclude those from our experiments since they do not contribute to defining a country’s mainstreamness.

3 Formalizing Mainstreamness

When describing how well a user’s listening preferences reflect those of an overall population, e.g., globally or within a country, what is considered *mainstream* depends on the selection of a population; this is a phenomenon which we will also show in our analysis. Consequently, we propose several quantitative measures for user mainstreamness, both on a global and on a country-specific level, depending on the selection of the population against which the target user is compared. Our approach is inspired by the well-established monotonicity assumptions in text processing and information

retrieval [37]: the TF-IDF (term frequency–inverse document frequency) weighting. Based on this assumption, our proposed mainstreamness measures rely on the concepts of *artist frequency* (AF), *listener frequency* (LF), and *artist frequency–inverse listener frequency* ($AF\cdot ILF$).

We define AF_{a,U_1} as the sum of the number of tracks by artist a listened to by a set of users U_1 . Note that U_1 may be a single user u , all users in a country c , or the entirety of users in the collection (i.e., the global population g). Accordingly, we define LF_{a,U_2} as the number of listeners of artist a within a user population U_2 . And we eventually define $AF\cdot ILF_{a,U_1,U_2}$ as in Equation 1. We set $AF\cdot ILF_{a,U_1,U_2} = 0$ iff $LF_{a,U_2} = 0$.

$$AF\cdot ILF_{a,U_1,U_2} = \log(1 + AF_{a,U_1}) \cdot \log\left(1 + \frac{|U_2|}{LF_{a,U_2}}\right) \quad (1)$$

Note that U_1 and U_2 may represent a single user, all users in the same country, or all users in the dataset (cf. Subsection 2.4). Therefore, this definition allows us to easily formalize both the global and the regional definitions of mainstreamness, by varying U_1 and U_2 . The ILF weighting term can be integrated when computing the *preference profile* for a user or for a country, e.g., $AF\cdot ILF_{a,u,c}$, where U_1 contains only the user u and U_2 all users in country c (to which u belongs), or $AF\cdot ILF_{a,c,g}$, where U_1 is composed of all users in country c (to which u belongs) and U_2 of all users in the dataset. Using ILF is motivated by the fact that, when determined by $AF_{a,c}$ or $LF_{a,c}$, the top artists in each country c are often identical or very similar to the global top artists (cf. Tables 1, 2, 3, and 4). In order to uncover the respective country-specific mainstream, we therefore use $ILF_{a,g}$ to penalize globally popular artists.

Table 1 Global top artists in the LFM-1b dataset, according to artist frequency (AF) and listener frequency (LF), considering the 53,258 users with country information

Artist	AF	Artist	LF
The Beatles	2,985,509	Radiohead	24,829
Radiohead	2,579,453	Nirvana	24,249
Pink Floyd	2,351,436	Coldplay	23,714
Metallica	1,970,569	Daft Punk	23,661
Muse	1,896,941	Red Hot Chili Peppers	22,609
Arctic Monkeys	1,803,975	Muse	22,429
Daft Punk	1,787,739	Queen	21,778
Coldplay	1,755,333	The Beatles	21,738
Linkin Park	1,691,122	Pink Floyd	21,129
Red Hot Chili Peppers	1,627,851	David Bowie	20,602

Table 2 Top artists for Finland (1,407 users), according to artist frequency (AF), listener frequency (LF), and artist frequency–inverse listener frequency (AF-ILF)

Artist	AF
Stam1na	105,633
In Flames	97,645
CMX	90,032
Kotiteollisuus	82,309
Turmion Katilot	78,722
Amorphis	78,159
Nightwish	75,742
Mokoma	73,453
Muse	69,507
Metallica	69,499
Artist	LF
Metallica	703
Nightwish	695
Muse	693
Daft Punk	675
Queen	671
System of a Down	663
Coldplay	634
Nirvana	614
Pendulum	613
Iron Maiden	609
Artist	AF-ILF
St. Hood	70.526
The Sun Sawed in 1/2	67.490
tiko-μ	66.546
Worth the Pain	66.058
Cutdown	65.247
Katariina Hänninen	64.955
Game Music Finland	64.835
Daisuke Ishiwatari	63.565
Altis	63.235
Redrum-187	62.428

Tables 2, 3, and 4 illustrate the effect of this weighting. It shows the top artists for Finland, Italy, and Turkey, in terms of $AF_{a,c}$, $LF_{a,c}$, and $AF \cdot ILF_{a,c,g}$, i.e., AF computed on the country level, ILF on the global level. As can be seen, the AF and even more the LF measures are not suited well to distill the essential mainstream of a country, except maybe for countries such as Finland that show a very specific music taste far away from the global

Table 3 Top artists for Italy (972 users), according to artist frequency (AF), listener frequency (LF), and artist frequency–inverse listener frequency (AF-ILF)

Artist	AF
Radiohead	68,160
The Beatles	65,498
Pink Floyd	60,558
Fabrizio De André	53,928
Muse	48,168
Depeche Mode	42,586
Afterhours	42,473
Verdena	42,338
Sigur Rós	41,748
Arctic Monkeys	39,755
Artist	LF
Radiohead	556
Pink Floyd	539
The Beatles	505
David Bowie	500
Muse	500
Nirvana	497
Coldplay	475
The Cure	466
Depeche Mode	459
Daft Punk	457
Artist	AF-ILF
CaneSecco	68.451
DSA Commando	66.049
Veronica Marchi	65.864
Train To Roots	65.459
Alessandro Raina	64.228
Machete Empire	63.915
Danti	62.958
Dargen D'Amico	62.453
宝塚歌劇団・宙組	62.228
Aquefrigide	61.663

taste [40]. In contrast, AF-ILF is capable of identifying those artists that are popular in a specific country, but not worldwide.

Based on the above definitions, we compute preference profiles globally (PP_g), for a country (PP_c), and for a user (PP_u). Given the LFM-1b dataset [39], these profiles are 585,095-dimensional vectors containing the AF, LF, or AF-ILF scores over all artists in the dataset. Figure 1 provides an example by visualizing the preference profiles for Finland, a country that does

Table 4 Top artists for Turkey (479 users), according to artist frequency (AF), listener frequency (LF), and artist frequency–inverse listener frequency (AF-ILF)

Artist	AF
Pink Floyd	68,887
Metallica	42,784
Daft Punk	42,020
Iron Maiden	34,174
Radiohead	31,390
Massive Attack	30,669
The Beatles	27,951
Opeth	25,744
Depeche Mode	25,075
Dream Theater	24,286
Artist	LF
Pink Floyd	292
Radiohead	289
Metallica	268
Coldplay	261
Nirvana	251
Massive Attack	249
The Beatles	240
Red Hot Chili Peppers	240
Queen	238
Led Zeppelin	236
Artist	AF-ILF
Cüneyt Ergün	64.473
Floyd Red Crow Westerman	61.955
Fırat Tanış	58.666
Acil Servis	58.439
Taste (Rory Gallagher)	58.366
Mezarkabul	57.799
Rachmaninoff Sergey	57.733
Mabel Matiz	57.619
Grup Yorum	56.855
Yüzyüzeyken Konuşuruz	56.748

particularly not correspond to the global music mainstream. Please note that artist IDs (on the x-axis) are sorted with respect to their *global* popularity in regards to the respective measure (AF, LF, or AF-ILF). As can be seen, while the distributions of the AF- and LF-based preference profiles follow a similar trend, the AF-ILF weighting considerably increases the importance of globally less popular, but country-wise more popular artists (also see Tables 2, 3, and 4).

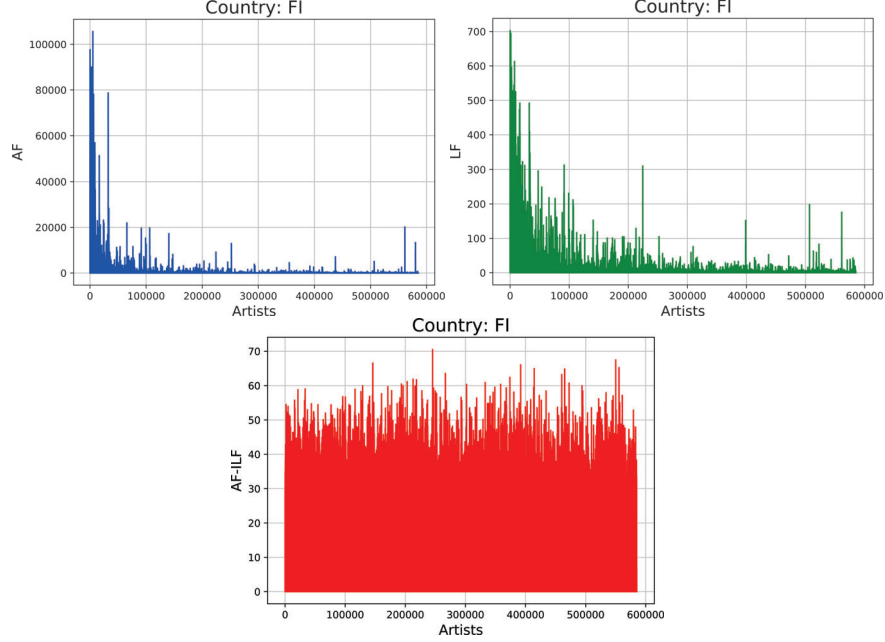


Figure 1 Artist frequency (AF), listener frequency (LF), and artist frequency–inverse listener frequency (AF-ILF) for Finland. Artist IDs (x-axis) are sorted by global AF, LF, or AF-ILF values, respectively.

Exploiting the profiles, we propose three categories of mainstreamness measures on the user level: fraction-based (F), symmetrized Kullback-Leibler divergence (D), and rank-order correlation according to Kendall’s τ (C). The adoption of fraction-based measures is motivated by their easy interpretability (due to the share of overlap between a user’s and the global or a country’s preference profiles). Kullback-Leibler divergence is a well-established method to compare distributions (discrete preference profiles in our case). Employing rank-order correlation is motivated by the fact that conversion of feature values to ranks has already been proven successful for music similarity tasks [32].

We provide formulas for the specific measures in Table 5, where $\hat{X}$ denotes the sum-to-unity normalized vector X and $\text{ranks}(PP_U^W)$ represents the real-valued preference profile converted to ranks, i.e. the vector containing all normalized item frequencies of user u , with respect to the frequency weighting approach W (AF or LF). When using AF-ILF, $\text{ranks}(PP_u^W)$ is extended to $\text{ranks}(PP_{u,c}^{AFILF})$, i.e. AF computed for user u , ILF on country c , or

Table 5 Proposed music mainstreamness measures on the user level. Terms denote the following: F stands for the fraction-based approach, D refers to the symmetrized Kullback-Leibler divergence approach, and C is used as abbreviation for the approaches based on rank-order correlation according to Kendall's τ . A is a list of all artists; $\widehat{AF}$ denotes the sum-to-unity normalized AF value; $ranks(PP_u^W)$ represents the real-valued preference profile converted to ranks, i.e. the vector containing all normalized item frequencies of user u , with respect to the frequency weighting approach W (AF or LF); in case of $AF \cdot ILF$, $ranks(PP_u^W)$ is extended to $ranks(PP_{u,c}^{AFILF})$, i.e. AF computed for user u , ILF on country c , or $ranks(PP_{c,g}^{AFILF})$, i.e. AF computed on country c , ILF globally. Note that we invert the values of some measures (F and D) in order to ensure that higher values always indicate closer to the mainstream

Abbr.	Formula
$F_{g:AF,u:AF}$	$1 - \frac{1}{ A } \cdot \sum_{a \in A} \frac{ \widehat{AF}_{a,u} - \widehat{AF}_{a,g} }{\max(\widehat{AF}_{a,u}, \widehat{AF}_{a,g})}$
$F_{g:AF,u:AF \cdot ILF}$	$1 - \frac{1}{ A } \cdot \sum_{a \in A} \frac{ AF \cdot \widehat{ILF}_{a,u,g} - \widehat{AF}_{a,g} }{\max(\widehat{AF} \cdot \widehat{ILF}_{a,u,g}, \widehat{AF}_{a,g})}$
$F_{g:AF \cdot ILF,u:AF \cdot ILF}$	$1 - \frac{1}{ A } \cdot \sum_{a \in A} \frac{ AF \cdot \widehat{ILF}_{a,u,g} - AF \cdot \widehat{ILF}_{a,g,g} }{\max(\widehat{AF} \cdot \widehat{ILF}_{a,u,g}, AF \cdot \widehat{ILF}_{a,g,g})}$
$F_{c:AF,u:AF}$	$1 - \frac{1}{ A } \cdot \sum_{a \in A} \frac{ \widehat{AF}_{a,u} - \widehat{AF}_{a,c} }{\max(\widehat{AF}_{a,u}, \widehat{AF}_{a,c})}$
$F_{c:AF \cdot ILF,u:AF \cdot ILF}$	$1 - \frac{1}{ A } \cdot \sum_{a \in A} \frac{ AF \cdot \widehat{ILF}_{a,u,c} - AF \cdot \widehat{ILF}_{a,c,g} }{\max(\widehat{AF} \cdot \widehat{ILF}_{a,u,c}, AF \cdot \widehat{ILF}_{a,c,g})}$
$D_{g:AF,u:AF}$	$\frac{1}{2} \cdot \left(\sum_{a \in A} \widehat{AF}_{a,u} \cdot \log \frac{\widehat{AF}_{a,u}}{\widehat{AF}_{a,g}} + \sum_{a \in A} \widehat{AF}_{a,g} \cdot \log \frac{\widehat{AF}_{a,g}}{\widehat{AF}_{a,u}} \right)^{-1}$
$D_{c:AF,u:AF}$	$\frac{1}{2} \cdot \left(\sum_{a \in A} \widehat{AF}_{a,u} \cdot \log \frac{\widehat{AF}_{a,u}}{\widehat{AF}_{a,c}} + \sum_{a \in A} \widehat{AF}_{a,c} \cdot \log \frac{\widehat{AF}_{a,c}}{\widehat{AF}_{a,u}} \right)^{-1}$
$D_{c:AF \cdot ILF,u:AF \cdot ILF}$	$\frac{1}{2} \cdot \left(\sum_{a \in A} AF \cdot \widehat{ILF}_{a,u,g} \cdot \log \frac{AF \cdot \widehat{ILF}_{a,u,g}}{AF \cdot \widehat{ILF}_{a,c,g}} + \sum_{a \in A} AF \cdot \widehat{ILF}_{a,c,g} \cdot \log \frac{AF \cdot \widehat{ILF}_{a,c,g}}{AF \cdot \widehat{ILF}_{a,u,g}} \right)^{-1}$
$C_{g:AF,u:AF}$	$\tau(ranks(PP_g^{AF}), ranks(PP_u^{AF}))$
$C_{c:AF,u:AF}$	$\tau(ranks(PP_c^{AF}), ranks(PP_u^{AF}))$
$C_{c:AF \cdot ILF,u:AF \cdot ILF}$	$\tau(ranks(PP_{u,c}^{AFILF}), ranks(PP_{c,g}^{AFILF}))$

$ranks(PP_{c,g}^{AFILF})$, i.e. AF computed on country c , ILF globally. Note that we invert the results of the fraction-based formulations and the symmetrized KL-divergences in order to be consistent in that higher values always indicate closer to the mainstream, while lower ones indicate farther away from the mainstream.

4 Analysis of Global Versus Country-Specific Mainstream

In order to identify archetypal countries for mainstreamness distributions, we investigate these distributions for the 47 countries in the dataset (cf. Subsection 2.4) that contain at least 100 listeners. Figure 2 illustrates four different examples, showing the country-specific listener frequency for the global top 50,000 artists, for the countries United States (US), Finland (FI), Brazil (BR), and Japan (JP). In all four plots, artists are sorted with respect to their global popularity in decreasing order along the x-axis. The black curve indicates the global trend, adjusted to the listener frequency in the respective country. Looking at the United States, we see that—except for some jitter—the distribution of listener frequencies among artists quite closely follows the global distribution (black curve). For Brazil, and even more for Finland, in contrast, a second trend curve becomes visible, indicating that in addition to the global trend (evidenced by a substantial amount of items along the black curve), certain artists within the countries are much more popular than expected from a global perspective. In Finland and Brazil, these country-specific popular artists follow approximately the same pattern as the global trend curve. In contrast, Japan does not reveal a clear secondary trend curve; there are rather many individual outliers that do not seem to follow a particular pattern.

To quantitatively identify and analyze the country-specific outliers that deviate from the global trend, we next use a sliding window of 5 artists, which we run over the top 1,000 AF, LF, and AF-ILF values of artists, sorted in the same way as in Figure 2, i.e., in decreasing order of global popularity, again for the top 47 countries in the dataset. We compute the mean AF, LF, and AF-ILF value within each window and relate it to the corresponding value of the first artist in the window. If this fraction exceeds a certain threshold, we consider the corresponding artist an outlier. For our experiments that we present in the following, we set that threshold to 100%, meaning that an outlier's value must be at least twice as large as the mean value in its window (in case of a positive outlier); or at most 50% of the value of the mean value in its window (in case of a negative outlier).

In doing so, we identify country-specific outliers that do not correspond to the global trend, meaning that the identified artists are particularly more (if positive) or particularly less popular in the respective country. Table 6 shows examples of positive AF outliers for Finland. Among the most salient outliers, we find the Finnish metal band “Amorphis”, but also metal bands from neighboring countries such as “Soilwork” from Sweden.

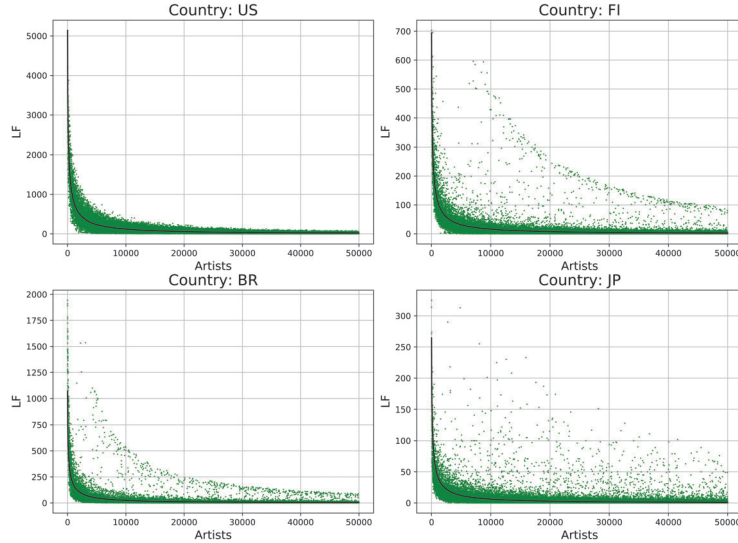


Figure 2 Country-specific listener frequency (LF) for global top 50,000 artists, for the United States (US), Finland (FI), Brazil (BR), and Japan (JP). In all four plots, artists are sorted with respect to their global popularity in decreasing order. The black curve indicates the global trend, adjusted to the LF in the respective country.

Table 6 Results of outlier analysis for artist–frequency (AF) values in Finland. The first 20 positive outliers are shown together with their global rank and the difference between their AF values and the mean AF values in a window of size 5, succeeding the artist

Artist	Rank	Difference
In Flames	25	+162.74%
Katatonia	73	+112.78%
Amon Amarth	90	+102.17%
Pendulum	99	+124.77%
Children of Bodom	122	+120.17%
Sonata Arctica	134	+146.35%
Bullet for My Valentine	138	+105.89%
HIM	154	+103.20%
Lamb of God	169	+136.27%
Sabaton	195	+168.01%
Amorphis	203	+229.48%
Infected Mushroom	220	+101.34%
Kamelot	248	+110.62%
Gojira	255	+128.40%
Dimmu Borgir	275	+140.08%
Soilwork	288	+220.73%
Burzum	305	+105.12%
Finntroll	314	+165.20%
Fear Factory	328	+122.30%
Biffy Clyro	365	+140.82%

Table 7 shows the top country-specific positive outliers for Germany. The artist with the highest AF difference to the expected AF values in its neighborhood (window) is “Die Ärzte”, a German punk rock band. Also other German bands rank high (e.g., “Rammstein”, “Volbeat”, and “In Extremo”).

To exemplify also negative outliers, Table 8 shows for the United States, the first (highest global position) positive and negative outliers that appear along the trend when using the AF measure. Among the negative outliers, we find mostly hard rock and metal bands, which corroborates previous findings that these genres are underrepresented in the United States compared to the global mean [42].

Table 7 Results of outlier analysis for artist–frequency (AF) values in Germany. The first 20 positive outliers are shown together with their global rank and the difference between their AF values and the mean AF values in a window of size 5, succeeding the artist

Artist	Rank	Difference
Rammstein	13	+115.87%
Rise Against	59	+128.29%
Mumford & Sons	85	+100.64%
Amon Amarth	90	+122.67%
Enter Shikari	179	+128.08%
Grateful Dead	261	+266.76%
Volbeat	287	+138.91%
3 Doors Down	298	+112.16%
Finntroll	314	+105.71%
Machine Head	325	+115.04%
The Gaslight Anthem	352	+102.57%
Biffy Clyro	365	+142.99%
Flogging Molly	395	+102.68%
Die Ärzte	437	+310.54%
Simple Plan	462	+158.99%
Heaven Shall Burn	505	+173.12%
La Dispute	541	+132.26%
Emilie Autumn	543	+116.91%
In Extremo	563	+194.80%
Combichrist	565	+121.34%

Table 8 Results of outlier analysis for artist–frequency (AF) values in the United States. The first 20 positive and negative outliers are shown together with their global rank and the difference between their AF values and the mean AF values in a window of size 5, succeeding the artist

Artist	Rank	Difference
Radiohead	1	+101.42%
Rammstein	13	−60.13%
Nine Inch Nails	20	+101.68%
Nightwish	23	−54.26%
In Flames	25	−54.56%
AC/DC	36	−53.89%
Korn	39	−53.46%
Marilyn Manson	52	−56.09%
The White Stripes	70	+112.77%
Katatonia	73	−60.63%
Within Temptation	74	−63.20%
30 Seconds to Mars	81	−56.39%
Guns N’ Roses	82	−63.45%
Amon Amarth	90	−55.56%
Anathema	97	−54.23%
Avenged Sevenfold	101	−64.63%
Modest Mouse	105	+142.16%
Bring Me the Horizon	106	−54.01%
Limp Bizkit	116	−73.35%
Blur	129	−54.05%

5 Music Recommendation Tailored to User Mainstreaminess

To evaluate the proposed mainstreaminess measures (cf. Section 3) with respect to their ability to improve performance in music recommendation, we conduct rating prediction experiments, which is a common approach to recommender systems evaluation. For this evaluation, we use again the LFM-1b dataset of user-generated listening events from Last.fm [39], as discussed in Subsection 2.4.

5.1 Experimental Setup

While we are aware that a truly user-centric evaluation would be beneficial for this kind of research, conducting a user study on tens of thousands of users (or even only a representative subset of the users) is beyond the scope of this paper. We therefore stick to the common approach of quantifying the performance of a recommender system by conducting a rating prediction task.

To this end, we normalize and scale the playcount values in the UAM to the range [0, 1000] for each user individually, assuming that higher numbers of playcounts indicate higher user preference for an artist.

We apply the common singular value decomposition (SVD) method according to [36] to factorize the UAM and in turn effect rating prediction. In 5-fold cross-validation experiments, we use root mean square error (RMSE) and mean absolute error (MAE) as performance measures.

To obtain a baseline, we first run the rating prediction experiment on the global group of 65,132 users and report results of the error measures in the first row of Table 9. To study the influence of both, the different mainstreamness

Table 9 Weighted root mean square error (RMSE) and weighted mean absolute error (MAE) for various mainstreamness definitions and levels, i.e. user sets. Rating values are scaled to [0, 1000]. Experiments are conducted on the country level (except for first row using the complete UAM with random item selection in each fold, irrespective of country) and error measures are averaged (arithmetic mean) over all countries with more than 1,000 users and weighted by number of users in the respective country. In the individual experiments, *all* refers to the group of all users in each considered country, *low* only to the users in the lower 3-quantile (tertile) w.r.t. the respective mainstreamness definition, *mid* and *high* defined analogously

Mainstreamness	User Set	w.RMSE	w.MAE
Baseline (global UAM)		29.105	25.202
$F_{g:AF,u:AF}$	<i>all</i>	26.377	24.050
	<i>high</i>	3.714	1.308
	<i>mid</i>	12.574	9.887
	<i>low</i>	14.186	11.625
$F_{g:AF,u:AF \cdot ILF}$	<i>all</i>	21.137	18.617
	<i>high</i>	3.681	1.299
	<i>mid</i>	11.035	8.191
	<i>low</i>	14.426	11.868
$F_{g:AF \cdot ILF,u:AF \cdot ILF}$	<i>all</i>	19.140	16.769
	<i>high</i>	11.777	9.121
	<i>mid</i>	13.396	10.833
	<i>low</i>	8.708	5.806
$F_{c:AF,u:AF}$	<i>all</i>	14.465	11.958
	<i>high</i>	3.723	1.309
	<i>mid</i>	8.681	6.112
	<i>low</i>	12.706	9.952
$F_{c:AF \cdot ILF,u:AF \cdot ILF}$	<i>all</i>	17.615	15.301
	<i>high</i>	9.237	6.648
	<i>mid</i>	3.686	1.305
	<i>low</i>	10.122	7.610

(Continued)

Table 9 Continued

Mainstreaminess	User Set	w.RMSE	w.MAE
$D_{g:AF,u:AF}$	<i>all</i>	24.026	21.705
	<i>high</i>	10.561	8.024
	<i>mid</i>	9.854	7.299
	<i>low</i>	5.365	2.909
$D_{c:AF,u:AF}$	<i>all</i>	28.021	25.746
	<i>high</i>	5.365	2.912
	<i>mid</i>	13.510	10.840
$D_{c:AF \cdot ILF, u:AF \cdot ILF}$	<i>all</i>	25.923	22.621
	<i>high</i>	14.628	11.624
	<i>mid</i>	3.656	1.281
	<i>low</i>	7.035	4.515
$C_{g:AF,u:AF}$	<i>all</i>	8.589	5.670
	<i>high</i>	15.906	13.525
	<i>mid</i>	3.680	1.291
	<i>low</i>	7.443	4.472
$C_{c:AF,u:AF}$	<i>all</i>	19.183	16.373
	<i>high</i>	14.349	12.032
	<i>mid</i>	3.687	1.290
$C_{c:AF \cdot ILF, u:AF \cdot ILF}$	<i>all</i>	4.270	1.833
	<i>high</i>	3.692	1.308
	<i>mid</i>	30.827	28.535
	<i>low</i>	7.680	5.187
	<i>all</i>	4.825	2.340
	<i>high</i>	10.785	8.1084
	<i>low</i>		

definitions and mainstreaminess *levels* on recommendation performance, we then create subsets of users for each combination of mainstreaminess measure and country with at least 1,000 users.¹ To this end, we split the users in each country into three (almost) equally sized subsets according to their mainstreaminess value: *low* corresponds to users in the lower 3-quantile (tertile) w.r.t. the respective mainstreaminess definition, *mid* and *high*, respectively, to the mid and upper tertile. In the individual experiments, *all* refers to the group of all users in each considered country, *low* only to the users in the lower 3-quantile (tertile) w.r.t. the respective mainstreaminess definition, *mid* and *high* defined analogously. Further, conducting the same experiment on all users in each country (user set *all*) allows for a comparison of a pure mainstreaminess filtering approach versus a combination of mainstreaminess filtering and demographic (country) filtering.

¹The restriction to countries with at least 1,000 users was made to allow for a meaningful analysis, as performed in [40].

5.2 Results and Discussion

Table 9 shows the error measures (RMSE and MAE) for different *definitions* and *levels* of mainstreamness, averaged over all considered countries (cf. Subsection 2.4), RMSE and MAE weighted by the number of users in the respective country. In the following discussion, we concentrate on RMSE since it is more common and considers larger differences between predicted and true ratings disproportionately more severe than smaller ones.

As a general finding, our results show that tailoring the recommendations to a user's mainstreamness level (*low*, *mid*, *high*) leads to substantial error reductions, irrespective of the applied mainstreamness measure. More specifically, $C_{c:AF,u:AF}$ outperforms the other measures in four regards: First, it leads to the lowest overall RMSE of 14.349 (*all*). Second, the errors realized by $C_{c:AF,u:AF}$ are also the lowest for each of the three user sets (*low*, *mid*, *high*). If better performance is achieved on a set with another measure, the difference is just in the third position after the decimal point. Third, $C_{c:AF,u:AF}$ performs on each of the three user sets (*low*, *mid*, *high*) in a balanced way (weighted RMSE amounts to respectively 3.692, 4.270, and 3.687), whereas the other mainstreamness measures yield a rather unbalanced picture since each of them performs on at least one set far worse than on the other(s), e.g., $C_{g:AF,u:AF}$ with 19.183, 7.443, and 3.681, respectively, for *low*, *mid*, and *high*. Fourth, $C_{c:AF,u:AF}$ performs well also on the low mainstreamness user set (*low*), which is a user segment that is typically difficult to satisfy.

The fraction-based approaches $F_{g:AF,u:AF}$, $F_{c:AF,u:AF}$, and $F_{g:AF,u:AF} \cdot ILF$ have in common that they perform far better in the high mainstreamness segment than in the mid and the low one. This could indicate that these measures still privilege globally popular items too much and, thus, produce more errors in the mid and low segments.

Interestingly, the approaches based on symmetrized Kullback-Leibler divergence (D) perform worse when tailored towards a user's country ($D_{c:AF,u:AF}$), compared to their application on a global level ($D_{g:AF,u:AF}$). Combining the country-specific tailoring with the AF-ILF weighting allows for better results compared to applying both separately.

While our results do not suggest a general superiority of mainstreamness measures that incorporate AF-ILF, first results of our deeper analysis on the country level indicate that these measures seem to perform particularly well for countries far from the global mainstream, such as Finland (RMSE of $D_{c:AF} \cdot ILF_{u:AF} \cdot ILF$ for *all*=5.985, *high*=1.346, *mid*=1.365, *low*=1.418), but worse for high mainstream countries, such as the USA (RMSE of

$D_{c:AF \cdot ILF, u:AF \cdot ILF}$ for $all=57.489$, $high=4.071$, $mid=4.077$, $low=55.968$). In the presented example, the low mainstream country Finland is small, and the respective weighted error measures in Table 9 do not reflect this country's users to the same extent as the large and high mainstream United States. As part of our ongoing large-scale analysis, delving into detail on country-specific aspects, we will investigate as a next step what factors influence the performance differences between countries for a given mainstreamness measure.

A direct comparison of the RMSE achieved by our approach with the RMSE reported in [48], the work closest to ours, is unfortunately impossible since Vigliensoni and Fujinaga quantized playcounts into a 5-point Likert rating scale: [1, 5]. Still, in a rough estimation, our results suggest that the accuracy of our best $C_{c:AF, u:AF}$ approach delivers a new benchmark in the combination of demographic (country) filtering and mainstreamness filtering, with a RMSE of 14.3 on a [0, 1000] scale. The best RMSE reported in [48] when considering mainstreamness and country information is approximately 0.9 on the much narrower [1, 5] scale (cf. approach *u.c.m.* in Figure 2 of [48]).

6 Conclusions and Outlook

The *music mainstreamness of a listener* reflects how strong a person's listening preferences correspond to those of the larger population. We consider that music mainstream may be defined from different perspectives. In this paper, we took into account that there are regional differences of what is considered mainstream, due to cultural characteristics and different market structures across countries.

The main contributions of this paper are three-fold: First, we proposed 11 novel measures to quantify the music mainstreamness of a user, a country, and an entire population. Those are based on fractional (F), divergence (D), and rank correlation (C) functions.

Second, we illustrated country-specific peculiarities of music preferences and *country-specific mainstream* employing the LFM-1b dataset [39]. We identified archetypal countries: (i) those countries where the mainstream of the country corresponds to the global trend (e.g., the United States), (ii) those countries with a distinct country-specific mainstream in addition to the global mainstream (e.g., Finland), and (iii) those countries roughly following the global mainstream trend without a clear secondary trend curve, but showing

various country-specific outliers over the whole global artist popularity range (e.g., Brazil and Japan).

Third, we studied the performance of the proposed mainstreamness measures for personalized music recommendation. Considering that music mainstream may be defined from a global but also a country-specific perspective, we particularly studied how the combination of a user’s mainstreamness and demographic (country) filtering influences the quality of music recommendations. Based on the LFM-1b dataset [39], we investigated the performance of the proposed measures in a rating prediction task, employing probabilistic matrix factorization. To quantify performance, we computed country-averaged, weighted RMSE and MAE figures for all mainstreamness definitions and various mainstreamness levels, and compared these with a global baseline. Overall, our results suggest that incorporating any kind of mainstreamness information outperforms the baseline. Our best approach combines demographic filtering (based on a user profile’s country) and mainstreamness filtering based on Kendall’s τ (variant $C_{c:AF,u:AF}$) and outperforms applying these filtering approaches separately. While our results do not hint at a general superiority of mainstreamness measures that incorporate AF-ILF, they do show that such measures perform much better than others for countries whose preference profiles are far away from the global taste (e.g., Finland).

As part of future work, we will take an in-depth look at the differences between countries, i.e. analyze in which countries which mainstreamness functions perform particularly well or poorly. Additionally, we plan to analyze how well our results generalize to other datasets providing demographic user information, e.g., the Million Musical Tweets Dataset [18], a playlist dataset crawled from Spotify users [31], or on a larger scale Spotify’s official Million Playlist Dataset,² released as part of the ACM Recommender Systems Challenge 2018 on automatic playlist continuation. We further plan user studies to investigate with qualitative methods whether incorporating mainstreamness information improves users’ perceived satisfaction with recommendations.

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²<https://recsys-challenge.spotify.com/details>

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